

Table 2. Alice-measured emission line brightness averages and SDs in sunlight and eclipse.

Emission line	Type	Disk-average brightness (rayleighs)		
		IEclipse01	IEclipse04	IEclipse05
OI 130.4 nm	Sunlight	706 ± 134	445 ± 27	347 ± 142
	Eclipse	597 ± 43	394 ± 14	254 ± 71
	Ratio S/E	1.18 ± 0.24	1.12 ± 0.08	1.37 ± 0.68
OI 135.6 nm	Sunlight	882 ± 177	577 ± 35	480 ± 188
	Eclipse	797 ± 57	536 ± 18	381 ± 94
	Ratio S/E	1.11 ± 0.24	1.08 ± 0.08	1.26 ± 0.58
SI 147.9 nm	Sunlight	1167 ± 271	986 ± 54	596 ± 288
	Eclipse	1205 ± 87	874 ± 28	429 ± 144
	Ratio S/E	0.97 ± 0.24	1.13 ± 0.07	1.39 ± 0.82

NH Long Range Reconnaissance Imager (LORRI) images (17). This same feature appears in the HST/SBC image in Fig. 2B, obtained when East Giru was shifted just behind the limb. The auroral feature near East Giru in both LORRI and HST/SBC images is ~15° northward of Jupiter’s field line tangent point at the limb, which suggests that ionospheric currents are diverted northward from this nominal position toward a region of higher gas density near the plume. Similar deviations of the anti-jovian FUV emissions from nominal tangent points observed with STIS are likely caused by the prevalence and distribution of plumes there (21). Volcanic plume aurorae were not identified in previous lower-quality STIS FUV images, which caused an apparent discrepancy with visible images of plume aurorae. The East Giru plume FUV auroral feature in Fig. 2 resolves this discrepancy and reveals the influence of plumes on Io’s electrodynamic interaction. The upstream-side emission feature is more apparent when limb brightened at viewing geometries like those reported in Fig. 2. This feature was predicted by aurora image simulations (22) and is diagnostic of the divergence of the plasma flow upstream of Io, a primary trait of Io’s interaction with the plasma torus.

References and Notes

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for emissions known to be located near the satellite disk (22); see, e.g., Fig. 2.
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REPORT

Io Volcanism Seen by New Horizons: A Major Eruption of the Tvashtar Volcano

J. R. Spencer,^{1*} S. A. Stern,² A. F. Cheng,² H. A. Weaver,³ D. C. Reuter,⁴ K. Retherford,⁵ A. Lunsford,⁴ J. M. Moore,⁶ O. Abramov,¹ R. M. C. Lopes,⁷ J. E. Perry,⁸ L. Kamp,⁷ M. Showalter,⁹ K. L. Jessup,¹ F. Marchis,⁹ P. M. Schenk,¹⁰ C. Dumas¹¹

Jupiter’s moon Io is known to host active volcanoes. In February and March 2007, the New Horizons spacecraft obtained a global snapshot of Io’s volcanism. A 350-kilometer-high volcanic plume was seen to emanate from the Tvashtar volcano (62°N, 122°W), and its motion was observed. The plume’s morphology and dynamics support nonballistic models of large Io plumes and also suggest that most visible plume particles condensed within the plume rather than being ejected from the source. In images taken in Jupiter eclipse, nonthermal visible-wavelength emission was seen from individual volcanoes near Io’s sub-Jupiter and anti-Jupiter points. Near-infrared emission from the brightest volcanoes indicates minimum magma temperatures in the 1150- to 1335-kelvin range, consistent with basaltic composition.

The New Horizons (NH) Jupiter flyby provided the first close-up observations of the tidally driven volcanism of Jupiter’s moon Io since the last Galileo orbiter observations of Io in late 2001 (1). The closest approach to Io occurred at 21:57 UT on 28 February 2007 at a range of 2.24 million km. Sunlit observations were made at solar phase angles from 5° to 159°, and four eclipses of Io by Jupiter were also observed. NH obtained 190 Io images with its 4.96 μrad per pixel panchromatic (400 to 900 nm) Long-Range Reconnaissance Imager (LORRI) and 17 color nighttime and eclipse images with the 20 μrad per pixel Multicolor Visible Imaging Camera (MVIC), although MVIC coverage of Io’s day side was not possible because of detector saturation. NH also obtained seven 1.25- to 2.5-μm near-infrared image cubes at 62 μrad per pixel with the Linear Etalon Infrared Spectral Array instrument

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Supporting Online Material

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Figs. S1 and S2
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